ACS 597: Noise Control Applications
Course wrap-up

Choose *one* of the problems below to complete. These problems are more open-ended than others in the course, and many of them combine concepts from multiple modules. You may modify the problems to suit your interests, or even make up an entirely new problem of similar scope to one of these. The goal is that you demonstrate your knowledge of some of the concepts and techniques you learned in this course. Feel free to send me a message with your idea if you are unsure if it meets these requirements or if you would like to discuss your approach.

I anticipate that 2–3 pages will be sufficient to describe your problem, approach, and results.

1 Woodwind or brass instrument modeling

Using techniques from the duct acoustics and sources module, model a different woodwind or brass instrument, such as a trumpet, trombone, tuba, clarinet, or flute. You should predict resonance frequencies with different configurations (e.g., trumpet valves in different positions) and make some comments (not necessarily numerical predictions) on the physics of the source driving the system.

2 Fan enclosure modeling

Read through the paper *Reduction of fan noise emission by enclosure modification*, posted to Canvas. Use the published dimensions and the authors' comments to calculate the resonance frequencies shown in Figure 4 of the paper. What causes the shift in resonances below 400 Hz? What do the other resonances correspond to?

The conclusions of the paper begin, "The effect of an enclosure on the sound radiation from fans mounted to it is expected to appear in various ways depending on the precise details of the situation." What are the "various ways" demonstrated in the paper? Cite the relevant figure in the paper for each effect.

3 DJ needle skipping

There are isolation pads and other products marketed to DJ's to prevent the vibrations of their turntables from causing the record needle to skip and the sound to become distorted. Model the turntable as a piece of sensitive equipment needing isolation. What are the potential sources of vibration a turntable might be subjected to? What do the forcing functions look like? Include at least a couple of different sources. Using products on the market or that you design, assess the performace of an isolation scheme on a turntable. Apart from isolation, what sorts of performance characteristics do you think a DJ might care about?

4 Sound mapping

A sound map, such as the National Transportation Noise Map, shows the predicted sound level over a given region due to a set of sources. Sound maps have been made for transportation noise, city noise, and wind farms. Create a sound map for sources and an area that interest you. You should include at least a couple of different sources operating simultaneously and include the source directivity and spreading to the far field.

5 Order tracking of mobius propeller data

Note: This problem may be best for those with prior signal analysis experience. The file `mobius_prop_data.mat` contains data from microphone array testing of a mobius propeller. Eleven microphones were placed to the side of the propeller as it ramped through RPM. Reproduce some of the order tracking plots from Lecture 26, which were using the first channel of this data set. Using all of the channels, can you see a difference in the directivity pattern at different orders? Note that the microphones are only spaced between -32 and 23 degrees, with 0 corresponding to perpendicular to the axis of rotation.